

Data Center Impact Explorer

Exploratory Data Analysis Presentation

Team JAMA

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Andrea MacGregor, Jillian Kunze, Anna Grace, and Morgan Murphy

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Project Goals

- ❖ We aim to investigate the relationships between data centers and local communities:
 - Data centers have a high water demand; are they located in areas already prone to drought, when they might pose additional strain?
 - Data centers also have high electricity demand; what are the conditions for electricity production and pricing where they are located?
 - Data centers produce pollution that can cause or exacerbate certain health conditions; what are the existing health concerns in local communities that could be aggravated?
 - Local governments play a role in data center siting in terms of zoning, permitting, and incentives; what are the political conditions in areas with data centers?

Data Manipulation, Cleaning, and Combining

Data Manipulation

Reformatting the data to be organized by ZIP/FIPS code-year or state-year (depending on combining need and when available)

Renaming columns to clearer names, using available data dictionaries

Data Cleaning

Removing null values, string manipulation, checking for duplicates, and data-type conversion

Paring down to useful and relevant columns

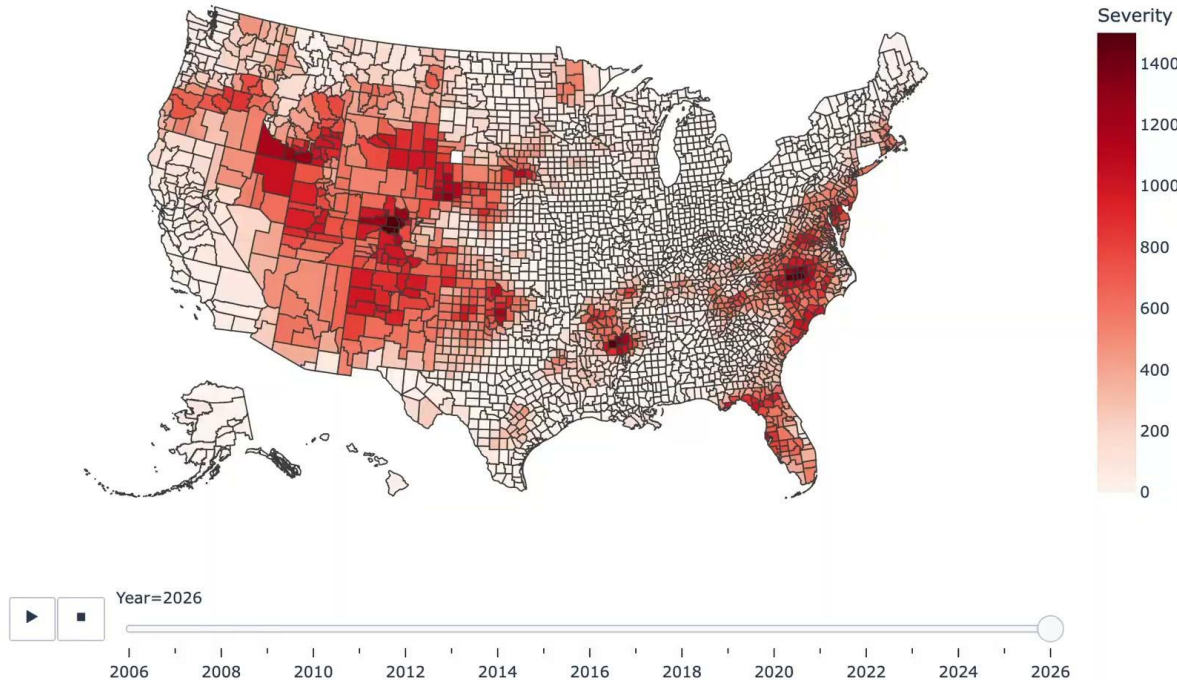
Data Combining and Aggregation

Developed process to convert from ZIP to FIPS code to enable data combining, and aggregate number of data centers per FIPS code

Combining datasets with data center locations

Environmental EDA

Annual Drought Severity (2006–2026)



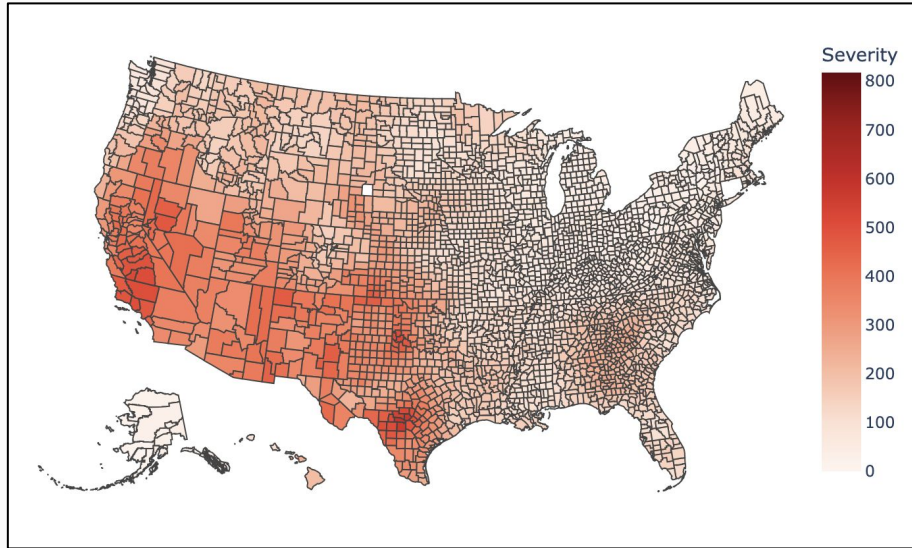
This is a snapshot of each year (one date per year)

Severity - Wanted to capture all drought on a single scale

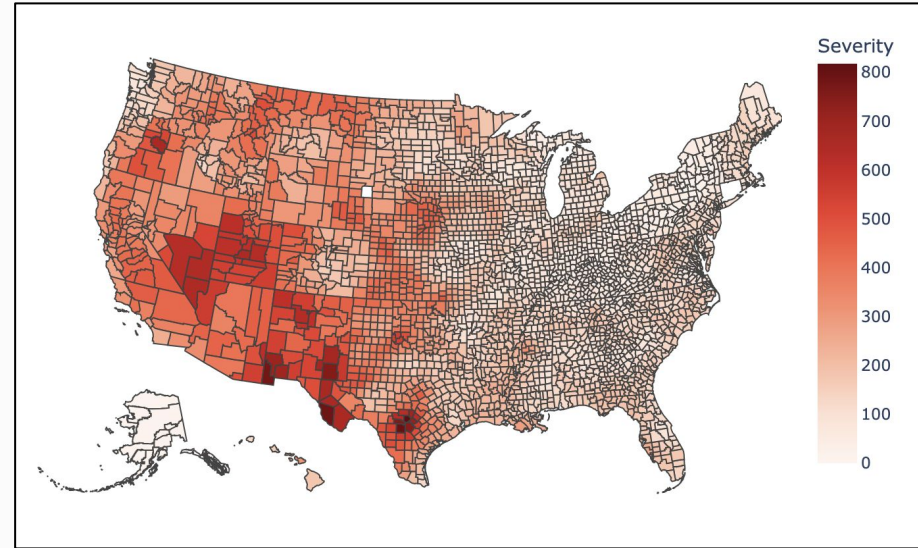
```
1 df["Severity"] = (  
2     1*df["Abnormally Dry"]  
3   + 2*df["Moderate Drought"]  
4   + 3*df["Severe Drought"]  
5   + 4*df["Extreme Drought"]  
6   + 5*df["Exceptional Drought"]  
7 )
```

Environmental EDA

Average Drought Severity (2016 - 2021)



Average Drought Severity (2022 - 2026)



- Average drought severity has increased from the previous 5 year period to the most recent 5 year period.
- Undetermined if the areas are correlated with data center locations.

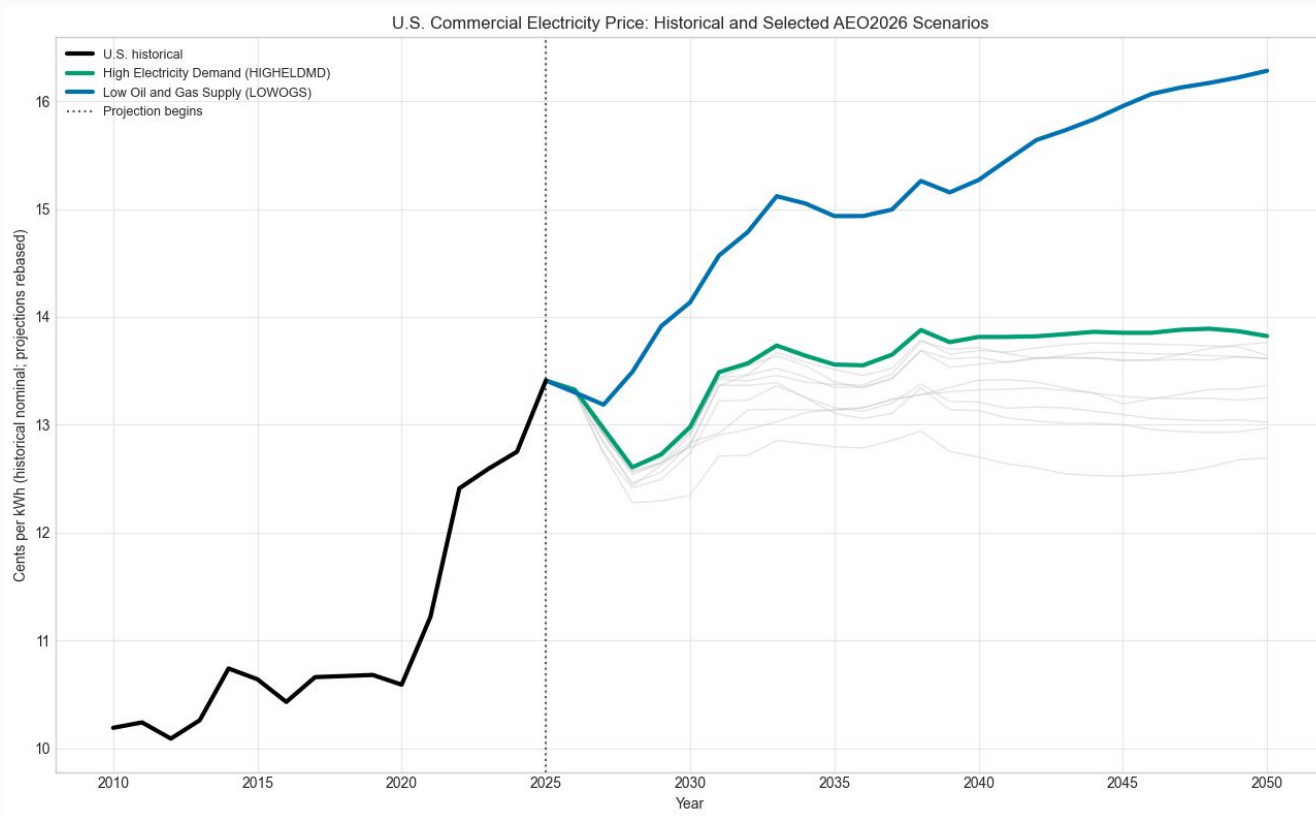
Energy EDA

EIA Annual Energy Outlook

- Project energy price and generation for the next 25 years
- Broken down into different scenarios

scenario_code	scenario
ALTELEC	Alternative Electricity
ALTTRNP	Alternative Transportation
CB2026	Counterfactual Baseline
ELECTRNP	Alternative Electricity and Alternative Transportation Combination
HIGHELMDM	High Electricity Demand
HIGHOGS	High Oil and Gas Supply
HIGHZTC	High Zero-carbon Technology Cost
HM2026	High Economic Growth
LM2026	Low Economic Growth
LOWOGS	Low Oil and Gas Supply
LOWZTC	Low Zero-carbon Technology Cost

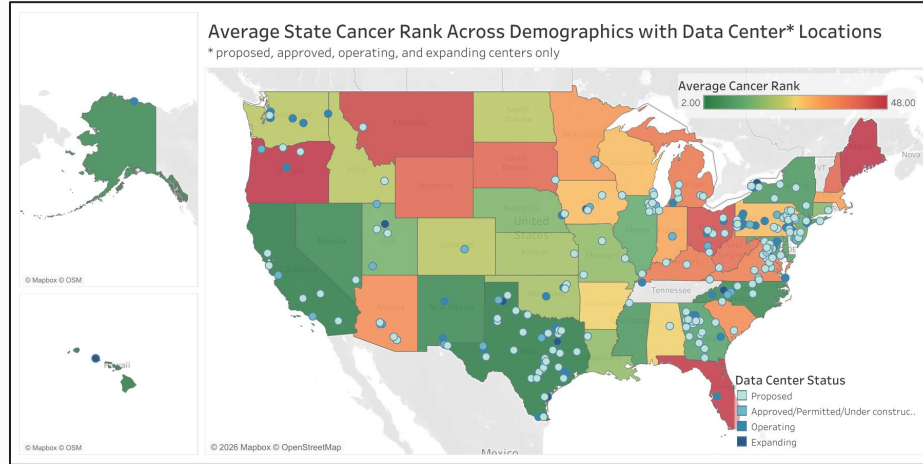
Energy EDA



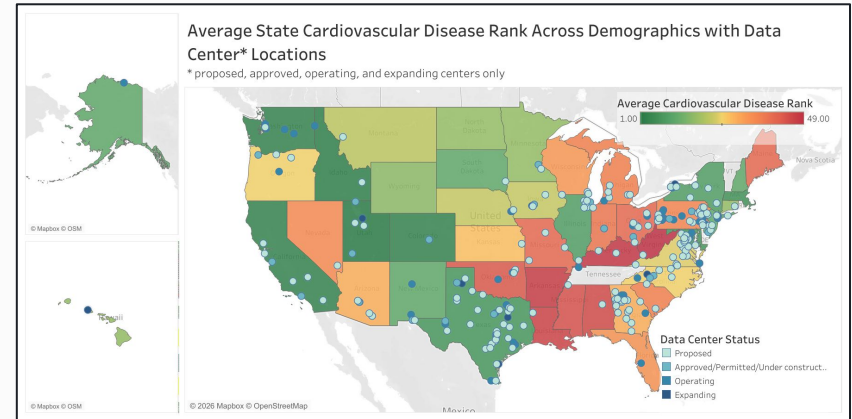
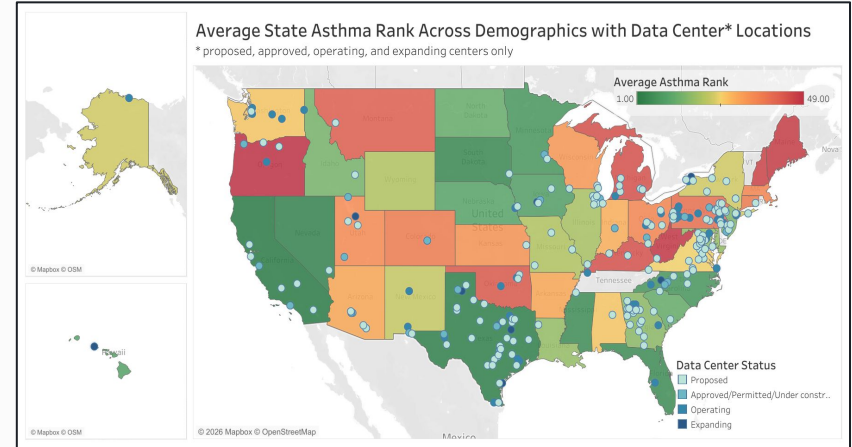
Commercial Electricity Pricing: Historical (2010-2025) and Projected (2026-2050),

The Low Oil and Gas Supply scenario shows the highest price per kWh on average, followed by High Energy Demand

Health EDA

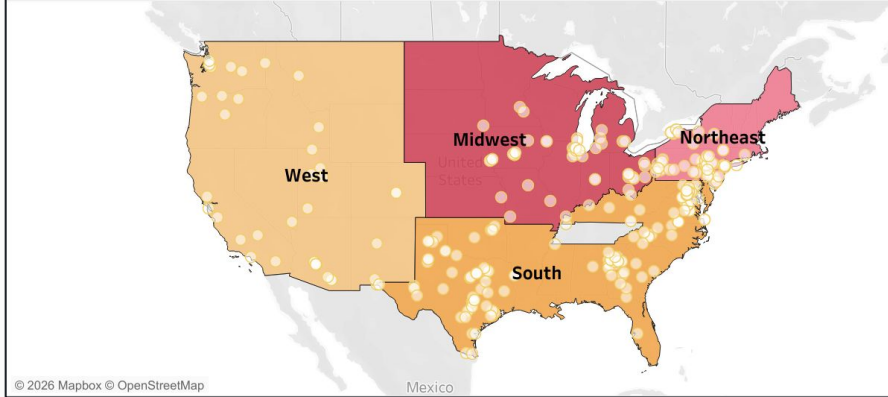


- Achieved by calculating the average asthma (**top right**), cancer (**left**), and cardiovascular disease (**bottom right**) rankings across all demographics per state
- Available on Tableau for visibility:
 - Class Discussion Board > “Form your group” Forum > “G1...” Thread

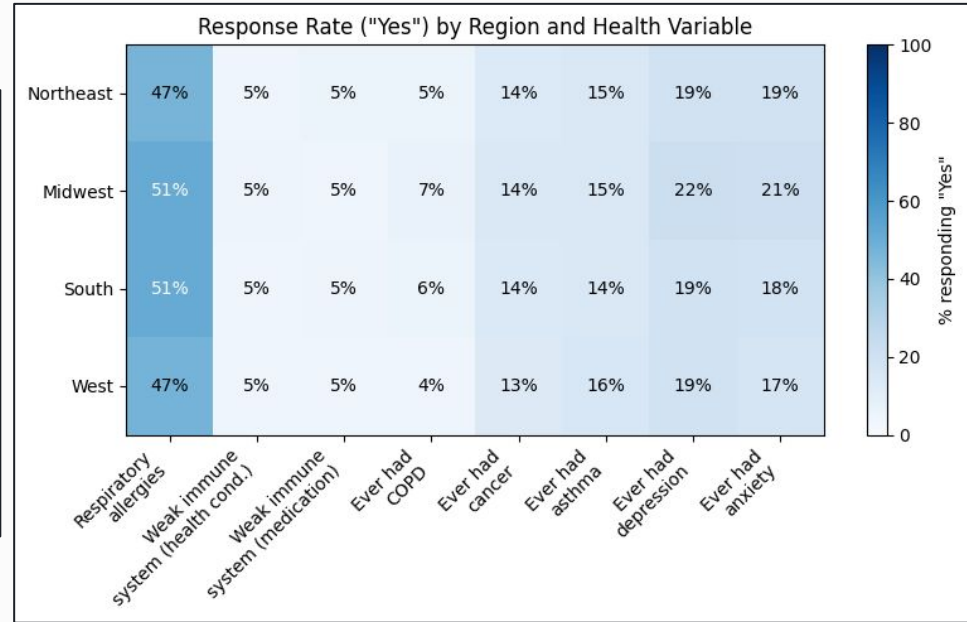


Health EDA

Proposed, Operating, or Expanding Data Centers by Geographic Region (lower 48)

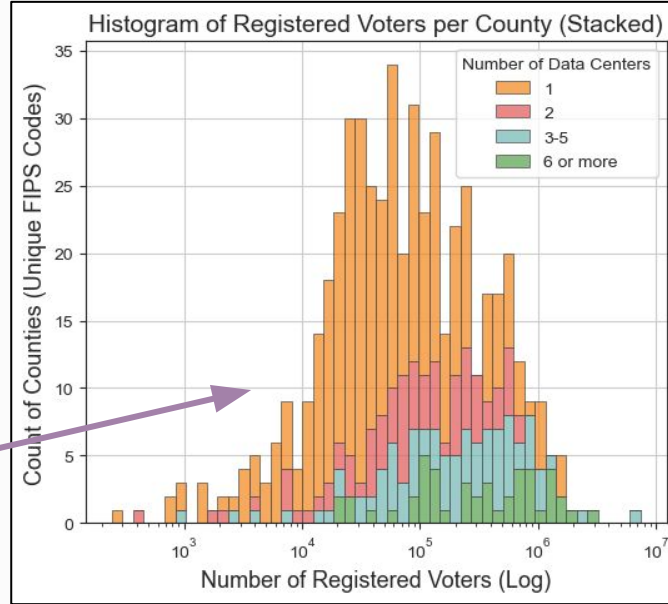
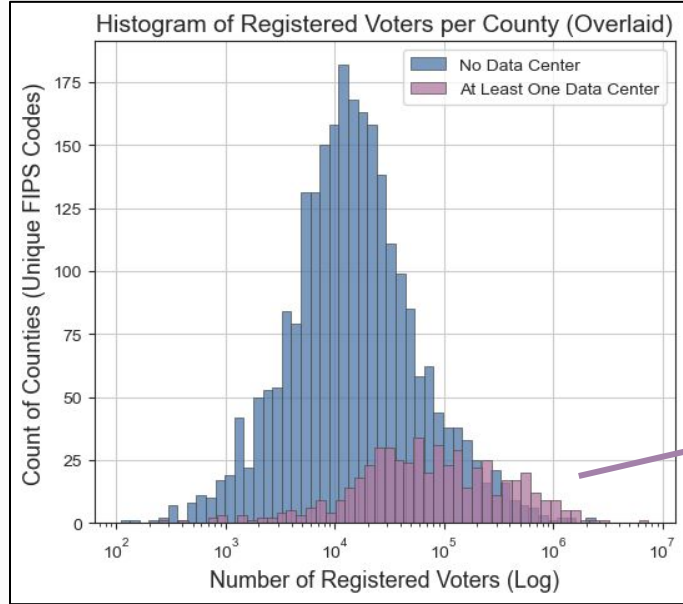


- **Top left:** states grouped by U.S. Census Bureau Geographic Region, in alignment with NHIS methodologies
- **Top right:** “Yes” response rate (%) heatmap for selected NHIS variables by region
- **Bottom right:** Chi-square and Cramér’s V scores for selected variables



CURRESP_A: $\chi^2=52.26$, $p=4.041e-08$, Cramér's $V=0.023$
HLTHCOND_A: $\chi^2=21.71$, $p=0.009853$, Cramér's $V=0.015$
MEDRXTRT_A: $\chi^2=14.65$, $p=0.1011$, Cramér's $V=0.012$
COPDEV_A: $\chi^2=44.27$, $p=1.258e-06$, Cramér's $V=0.021$
CANEV_A: $\chi^2=10.62$, $p=0.3029$, Cramér's $V=0.010$
ASEV_A: $\chi^2=15.51$, $p=0.07795$, Cramér's $V=0.013$
DEPEV_A: $\chi^2=44.40$, $p=1.189e-06$, Cramér's $V=0.021$
ANXEV_A: $\chi^2=61.04$, $p=8.453e-10$, Cramér's $V=0.025$

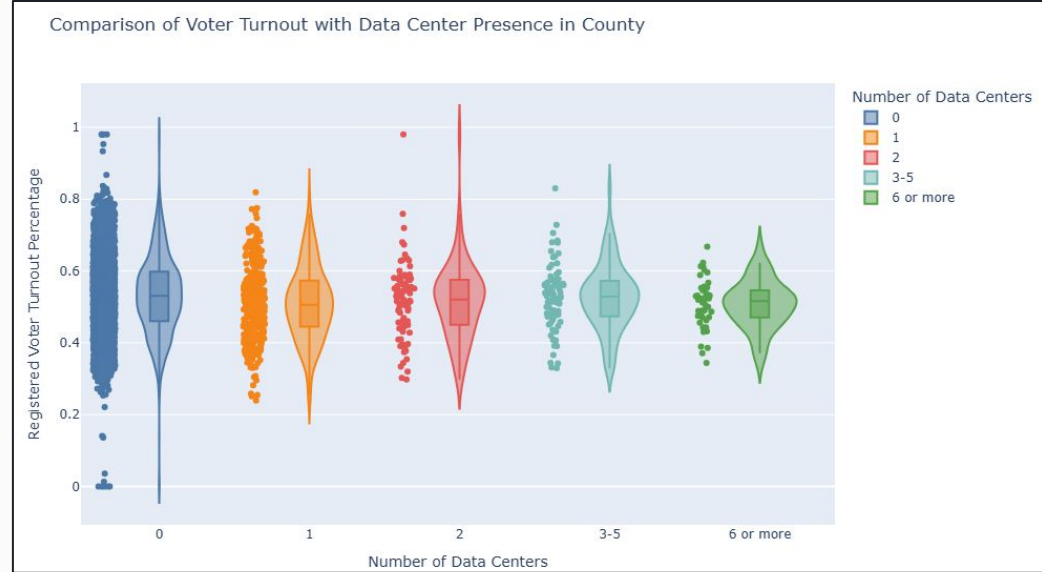
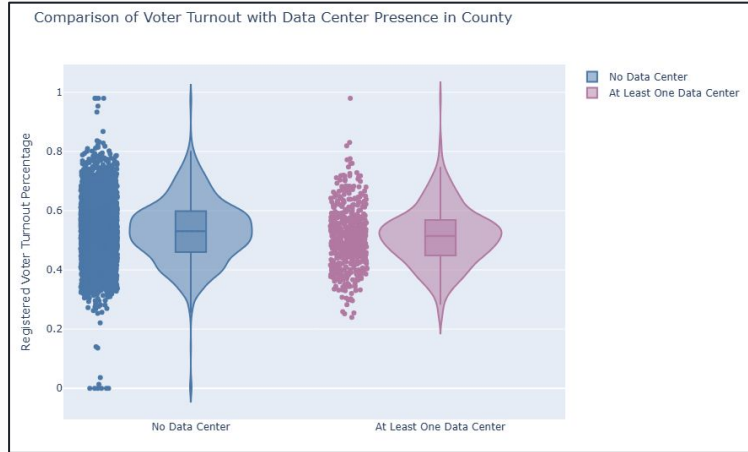
Political EDA



Number of Data Centers	Mean Number of Registered Voters
0	41,108
1	129,861
2	166,535
3-5	422,430
6 or more	564,258

Data from FracTrackers U.S. Data Centers Tracker and National Neighborhood Data Archive

Political EDA



Number of Data Centers	Mean Registered Voter Turnout Percentage
0	53%
1	51%
2	51%
3-5	53%
6 or more	51%

Data from FracTrackers U.S. Data Centers Tracker and National Neighborhood Data Archive

Initial investigation shows much greater differences for political partisanship (stay tuned!)

Next Steps

Environmental

- Average each year for animated graph
- Convert ZIP to FIPS codes
- Add data center locations to graphs

Energy

- Work on state-level projections
- Map energy price/generation (similar to drought)

Health

- Continue integrating state and region diagnosis data
- Map health data on the regional level

Political

- Visualize partisanship in federal voting
- Find trends in locally elected officials and combine with data center data

- Create a consolidated interactive public dashboard using Streamlit



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Questions?